CHAPTER 12
multi-asset options

The aim of this Chapter...

...is to introduce the idea of correlation between many different assets and so develop a theory for derivatives that depend on several different assets simultaneously.

In this Chapter...

- how to model the behavior of many assets simultaneously
- estimating correlation between asset price movements
- how to value and hedge options on many underlying assets in the Black–Scholes framework
- the pricing formula for European non-path-dependent options on dividend-paying assets

12.1 INTRODUCTION

In this chapter I introduce the idea of higher dimensionality by describing the Black–Scholes theory for options on more than one underlying asset. This theory is perfectly straightforward; the only new idea is that of correlated random walks and the corresponding multifactor version of Itô's lemma.

Although the modeling and mathematics is easy, the final step of the pricing and hedging, the 'solution,' can be extremely hard indeed. I explain what makes a problem easy, and what makes it hard, from the numerical analysis point of view.

12.2 MULTIDIMENSIONAL LOGNORMAL RANDOM WALKS

The basic building block for option pricing with one underlying is the lognormal random walk

$$dS = \mu S\,dt + \sigma S\,dX.$$

This is readily extended to a world containing many assets via models for each underlying

$$dS_i = \mu_i S_i\,dt + \sigma_i S_i\,dX.$$

Here S_i is the price of the ith asset, $i = 1, \dots, d$, and μ_i and σ_i are the drift and volatility of that asset respectively and dX_i is the increment of a Wiener process. We can still continue to think of dX_i as a random number drawn from a Normal distribution with mean zero and standard deviation $dt^{1/2}$ so that

$$E[dX_i] = 0 \quad \text{and} \quad E[dX_i^2] = dt$$

but the random numbers dX_i and dX_j are **correlated**:

$$E[dX_i\,dX_j] = \rho_{ij}\,dt.$$

here ρ_{ij} is the correlation coefficient between the ith and jth random walks. The symmetric matrix with ρ_{ij} as the entry in the ith row and jth column is called the **correlation matrix**. For example, if we have seven underlyings $d = 7$ and the correlation matrix will look like this:

$$\mathbf{\Sigma} = \begin{pmatrix} 1 & \rho_{12} & \rho_{13} & \rho_{14} & \rho_{15} & \rho_{16} & \rho_{17} \\ \rho_{21} & 1 & \rho_{23} & \rho_{24} & \rho_{25} & \rho_{26} & \rho_{27} \\ \rho_{31} & \rho_{32} & 1 & \rho_{34} & \rho_{35} & \rho_{36} & \rho_{37} \\ \rho_{41} & \rho_{42} & \rho_{43} & 1 & \rho_{45} & \rho_{46} & \rho_{47} \\ \rho_{51} & \rho_{52} & \rho_{53} & \rho_{54} & 1 & \rho_{56} & \rho_{57} \\ \rho_{61} & \rho_{62} & \rho_{63} & \rho_{64} & \rho_{65} & 1 & \rho_{67} \\ \rho_{71} & \rho_{72} & \rho_{73} & \rho_{74} & \rho_{75} & \rho_{76} & 1 \end{pmatrix}$$

Note that $\rho_{ii} = 1$ and $\rho_{ij} = \rho_{ji}$. The correlation matrix is positive definite, so that $\mathbf{y}^T\mathbf{\Sigma}\mathbf{y} \geq 0$. The **covariance matrix** is simply

$$\mathbf{M\Sigma M},$$

where $\mathbf{M}$ is the matrix with the σ_i along the diagonal and zeros everywhere else.

To be able to manipulate functions of many random variables we need a multidimensional version of Itô's lemma. If we have a function of the variables $S_1, \dots, S_d$ and t, $V(S_1, \dots, S_d, t)$, then

$$dV = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sum_{i=1}^{d} \sum_{j=1}^{d} \sigma_i \sigma_j \rho_{ij} S_i S_j \frac{\partial^2 V}{\partial S_i \, \partial S_j} \right) dt + \sum_{i=1}^{d} \frac{\partial V}{\partial S_i} dS_i.$$

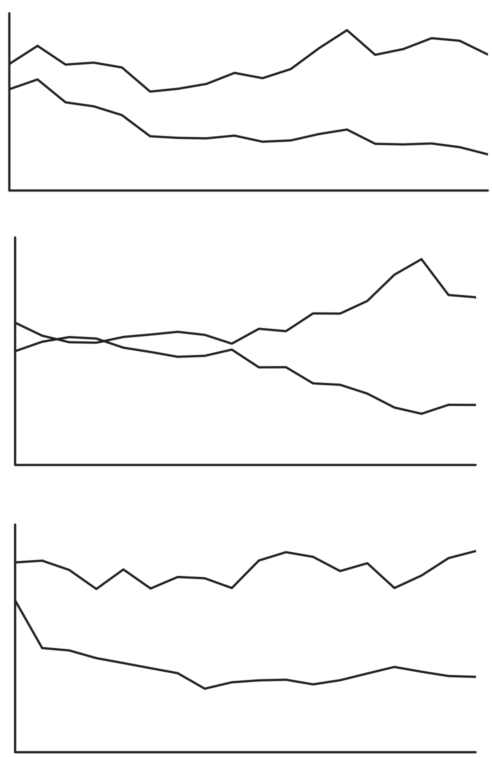
We can get to this same result by using Taylor series and the rules of thumb:

$$dX_i^2 = dt \quad \text{and} \quad dX_i dX_j = \rho_{ij} \, dt.$$

Time Out...

Correlation

Correlation is a measure of the relationship or dependence between two or more random quantities.



Excel simulation

This is most easily explained by reference to the above series of figures. In the first figure we see two random walks that are perfectly correlated, $\rho = 1$. They may be moving apart overall, but that is a long-term phenomenon. In the short term, and correlation is a characteristic of random walks over small periods of time, you can see that each up move in one random walk is matched by an up move in the other.

In the second figure we see a correlation ρ of -1. Now each up move in one walk is matched by a down in the other. The third figure shows two uncorrelated random walks, there is no relationship between the up and down moves in the two walks.

The correlation can be anywhere between -1 and $+1$. What would two random walks with a correlation of 0.5 look like?

P.S. I don't believe in correlations among financial assets.

12.3 MEASURING CORRELATIONS

If you have time series data at intervals of δt for all d assets you can calculate the correlation between the returns as follows. First, take the price series for each asset and calculate the return over each period. The return on the ith asset at the kth data point in the time series is simply

$$R_i(t_k) = \frac{S_i(t_k + \delta t) - S_i(t_k)}{S_i(t_k)}.$$

The historical volatility of the ith asset is

$$\sigma_i = \sqrt{\frac{1}{\delta t(M-1)} \sum_{k=1}^{M} (R_i(t_k) - \overline{R}_i)^2}$$

where M is the number of data points in the return series and $\overline{R}_i$ is the mean of all the returns in the series.

The covariance between the returns on assets i and j is given by

$$\frac{1}{\delta t(M-1)} \sum_{k=1}^{M} (R_i(t_k) - \overline{R}_i)(R_j(t_k) - \overline{R}_j).$$

The correlation is then

$$\frac{1}{\delta t(M-1)\sigma_i\sigma_j} \sum_{k=1}^{M} (R_i(t_k) - \overline{R}_i)(R_j(t_k) - \overline{R}_j).$$

In Excel correlation between two time series can be found using the CORREL worksheet function, or Tools | Data Analysis | Correlation.

Figure 12.1 shows the correlation matrix for Marks & Spencer, Tesco, Sainsbury and IBM.

Correlations measured from financial time series data are notoriously unstable. If you split your data into two equal groups, up to one date and beyond that date, and calculate the correlations for each group you may find that they differ quite markedly. You could calculate a 60-day correlation, say, from several years' data and the result would look something like Figure 12.2. You might want to use a historical 60-day correlation if you have a contract of that maturity. But, as can be seen from the figure, such a historical correlation should be used with care; correlations are even more unstable than volatilities.

The other possibility is to back out an **implied correlation** from the quoted price of an instrument. The idea behind that approach is the same as with implied volatility, it gives an estimate of the market's perception of correlation.

Time Out...

On a spreadsheet

The following spreadsheet shows how to simulate two correlated random walks on a spreadsheet. Both of these random walks are lognormal, but notice the correlation between them.

	A	B	C	D	E	F	G	H	I
1	**Asset1**	**Asset2**		**Time**	**Random1**	**Random2**	**Asset1**	**Asset2**	
2	**100**	**80**		0	0.046223	–1.59903	100	80	
3				0.01	–0.158143	–0.960557	99.78371	77.97375	
4	**Drift1**	**Drift2**		0.02	–0.540749	0.340648	98.80434	78.18732	
5	**0.1**	**0.2**		.03	0.859933	–1.754755	100.6024	75.78769	
6				0.04	–0.268174	0.896078	100.1635	77.3988	
7	**Vol1**	**Vol2**		0.05	–0.810562	–2.361049	98.63986	71.86476	
8	**0.2**	**0.3**		6	–0.974247	0.569597	96.81651	72.02177	
9				7	0.576045	1.016849	98.02874	74.69084	
10	**Correl.**	**0.5**		8	–0.989892	–0.409346	96.18601	72.93684	
11				9	–0.839252	–1.013799	94.66771	70.24342	
12	**Timestep**	**0.01**		1	0.372974	–0.409777	95.46855	70.02906	
13				1	–0.542291	–0.597359	94.52858	68.51264	
14	**Sqrt(1-correl^2)**			2	0.248432	–0.643216	95.09279	67.76004	
15	0.866025			3	0.963828	1.237832	97.02094	71.05435	
16				.14	0.591412	2.049829	98.26555	75.61087	
17				0.15	–0.243018	0.321826	97.88621	76.11868	
18				0.16	0.558761	1.187003	99.078	79.25634	
19				0.17	–0.951554	1.985109	97.29151	82.37122	
20				0.18	0.502183	–1.61082	98.36597	79.70918	
21					72	–1.502543	95.92222	75.212	
22				0.2	0.348241	–1.424413	96.68622	72.97191	
23				0.21	2.09392	–1.355961	100.832	72.8391	
24				0.22	1.044282	–1.60787	103.0387	71.08298	
25				0.23	0.902542	1.108671	105.0017	74.23496	
26				0.24	1.517429	–1.260644	108.2934	73.64175	
27							8.629	75.89836	
28				0.26	–0.640458	0.532359	107.3462	76.37077	
29				0.27	0.344599	–0.17648	108.1934	76.5681	
30				0.28	1.082126	0.483641	110.6431	78.92619	
31				0.29	0.041153	–0.865378	110.8448	77.35825	
32				0.3	–0.029683	0.034647	110.8899	77.54816	

= D3 + B12

= RAND() + RAND() + RAND() + RAND() + RAND() + RAND() + RAND() + RAND() + RAND() + RAND() + RAND() + RAND() –6

= SQRT(1–B10*B10)

= G19*(1 + A$5*$B$12 + A$8*SQRT(B12)*E20)

= H24*(1 + B$5*$B$12 + B$8*SQRT(B12)*(B10*E25 + A15*F25))

Excel simulation

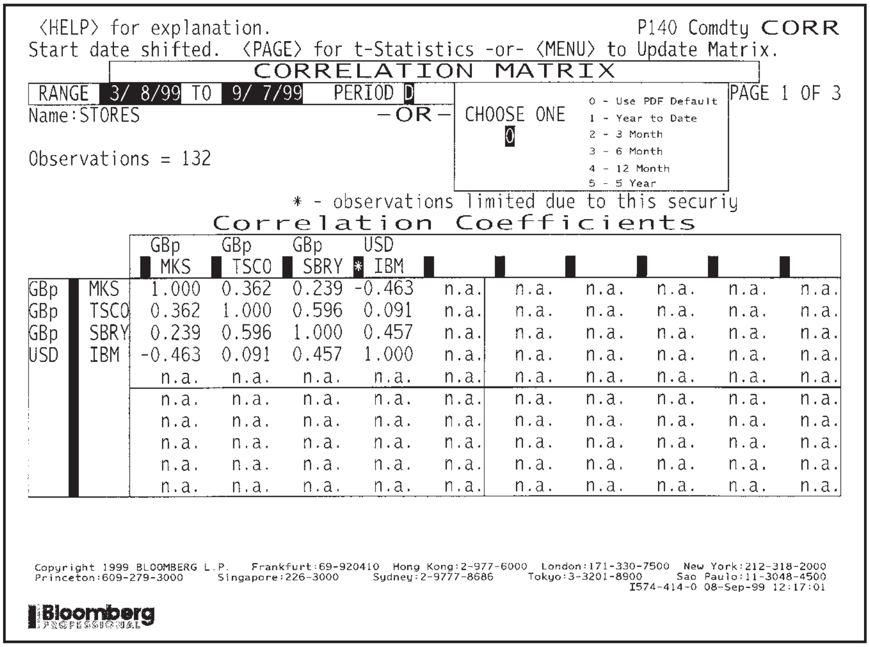


Figure 12.1 Some correlations. Source: Bloomberg L.P.

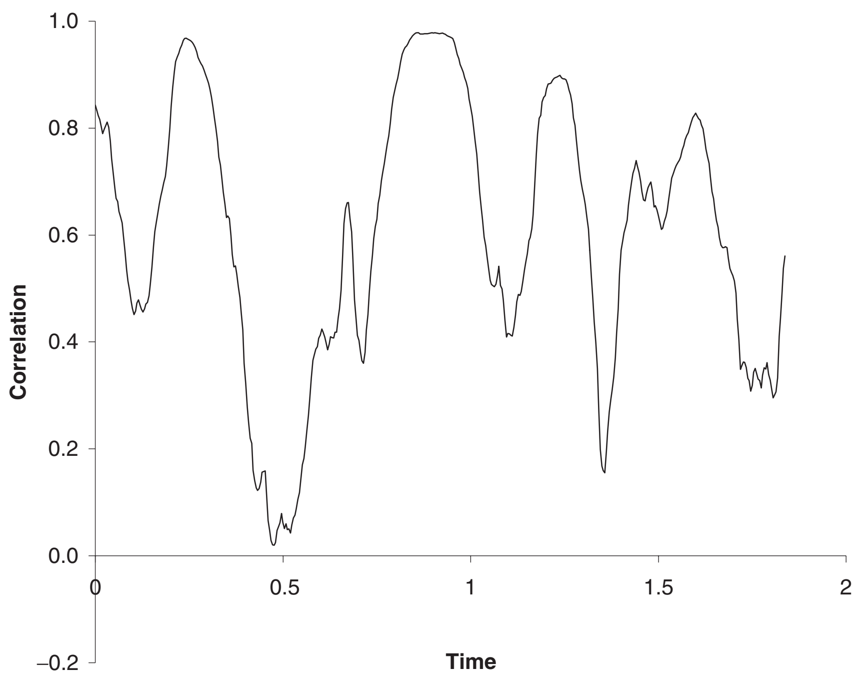


Figure 12.2 A correlation time series.

12.4 OPTIONS ON MANY UNDERLYINGS

Options with many underlyings are called **basket options**, **options on baskets** or **rainbow options**. The theoretical side of pricing and hedging is straightforward, following the Black–Scholes arguments but now in higher dimensions.

Set up a portfolio consisting of one basket option and short a number Δ_i of each of the assets S_i:

$$\Pi = V(S_1, \dots, S_d, t) - \sum_{i=1}^{d} \Delta_i S_i.$$

The change in this portfolio is given by

$$d\Pi = \left(\frac{\partial V}{\partial t} + \tfrac{1}{2} \sum_{i=1}^{d} \sum_{j=1}^{d} \sigma_i \sigma_j \rho_{ij} S_i S_j \frac{\partial^2 V}{\partial S_i \, \partial S_j} \right) dt + \sum_{i=1}^{d} \left(\frac{\partial V}{\partial S_i} - \Delta_i \right) dS_i.$$

If we choose

$$\Delta_i = \frac{\partial V}{\partial S_i}$$

for each i, then the portfolio is hedged and is risk-free. Setting the return equal to the risk-free rate we arrive at

$$\frac{\partial V}{\partial t} + \tfrac{1}{2} \sum_{i=1}^{d} \sum_{j=1}^{d} \sigma_i \sigma_j \rho_{ij} S_i S_j \frac{\partial^2 V}{\partial S_i \, \partial S_j} + r \sum_{i=1}^{d} S_i \frac{\partial V}{\partial S_i} - rV = 0. \qquad (12.1)$$

This is the multidimensional version of the Black–Scholes equation. The modifications that need to be made for dividends are obvious. When there is a dividend yield of D_i on the ith asset we have

$$\boxed{\frac{\partial V}{\partial t} + \tfrac{1}{2} \sum_{i=1}^{d} \sum_{j=1}^{d} \sigma_i \sigma_j \rho_{ij} S_i S_j \frac{\partial^2 V}{\partial S_i \, \partial S_j} + \sum_{i=1}^{d} (r - D_i) S_i \frac{\partial V}{\partial S_i} - rV = 0}$$

Time Out...

Here we go again

Risk neutrality means that the drift rates of the assets do not appear in the pricing equation.

12.5 THE PRICING FORMULA FOR EUROPEAN NON-PATH-DEPENDENT OPTIONS ON DIVIDEND-PAYING ASSETS

Because there is a Green's function for this problem (see Chapter 7) we can write down the value of a European non-path-dependent option with payoff of Payoff($S_1, \dots, S_d$) at time T:

$$V = e^{-r(T-t)}\left(2\pi(T-t)\right)^{-d/2}(\mathrm{Det}\boldsymbol{\Sigma})^{-1/2}(\sigma_1\dots\sigma_d)^{-1}$$
$$\int_0^\infty \cdots \int_0^\infty \frac{\mathrm{Payoff}(S_1'\cdots S_d')}{S_1'\dots S_d'}\exp\left(-\frac{1}{2}\boldsymbol{\alpha}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\alpha}\right)dS_1'\cdots dS_d'. \tag{12.2}$$
$$\alpha_i = \frac{1}{\sigma_i(T-t)^{1/2}}\left(\log\left(\frac{S_i}{S_i'}\right) + \left(r - D_i - \frac{\sigma_i^2}{2}\right)(T-t)\right)$$

This has included a constant continuous dividend yield of D_i on each asset.

Time Out...

Ouch! But don't worry

This formula looks horrible. But it has a simple interpretation as the present value of an expectation of the payoff. Part of the integrand (the bit inside the integral) is the payoff, and part represents the probability density function. The numerical integration of this expression is actually quite straightforward as we'll see in Chapter 30.

As always, it's the risk-neutral expectation that matters. Do you see any μs anywhere in the formula? No.

12.6 EXCHANGING ONE ASSET FOR ANOTHER: A SIMILARITY SOLUTION

An **exchange option** gives the holder the right to exchange one asset for another, in some ratio. The payoff for this contract at expiry is

$$\max(q_1S_1 - q_2S_2, 0),$$

where q_1 and q_2 are constants.

The partial differential equation satisfied by this option in a Black–Scholes world is

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sum_{i=1}^{2}\sum_{j=1}^{2}\sigma_i\sigma_j\rho_{ij}S_iS_j\frac{\partial^2 V}{\partial S_i\,\partial S_j} + \sum_{i=1}^{2}(r - D_i)S_i\frac{\partial V}{\partial S_i} - rV = 0.$$

A dividend yield has been included for both assets. Since there are only two underlyings the summations in these only go up to two.

This contract is special in that there is a similarity reduction. Let's postulate that the solution takes the form

$$V(S_1, S_2, t) = q_1 S_2 H(\xi, t),$$

where the new variable is

$$\xi = \frac{S_1}{S_2}.$$

If this is the case, then instead of finding a function V of three variables, we only need find a function H of two variables, a much easier task.

Time Out...

Similarity reductions

If you skipped much of Chapter 7, as I advised some of you to do, you won't have read about similarity reductions. This is just a useful trick, not one you can often use, but when you can, you should.

Sometimes it is possible to reduce the number of dimensions in a problem by exploiting the 'nice' form of the problem. The example here is typical.

Changing variables from S_1, S_2 to ξ we must use the following for the derivatives.

$$\frac{\partial}{\partial S_1} = \frac{1}{S_2}\frac{\partial}{\partial \xi}, \quad \frac{\partial}{\partial S_2} = -\frac{\xi}{S_2}\frac{\partial}{\partial \xi},$$

$$\frac{\partial^2}{\partial S_1^2} = \frac{1}{S_2^2}\frac{\partial^2}{\partial \xi^2}, \quad \frac{\partial^2}{\partial S_2^2} = \frac{\xi^2}{S_2^2}\frac{\partial^2}{\partial \xi^2} + \frac{2\xi}{S_2^2}\frac{\partial}{\partial \xi}, \quad \frac{\partial^2}{\partial S_1\,\partial S_2} = -\frac{\xi}{S_2^2}\frac{\partial^2}{\partial \xi^2} - \frac{1}{S_2^2}\frac{\partial}{\partial \xi}.$$

The time derivative is unchanged. The partial differential equation now becomes

$$\frac{\partial H}{\partial t} + \frac{1}{2}\sigma'^2\xi^2\frac{\partial^2 H}{\partial \xi^2} + (D_2 - D_1)\xi\frac{\partial H}{\partial \xi} - D_2 H = 0.$$

where

$$\sigma' = \sqrt{\sigma_1^2 - 2\rho_{12}\sigma_1\sigma_2 + \sigma_2^2}.$$

You will recognize this equation as being the Black–Scholes equation for a single stock with D_2 in place of r, D_1 in place of the dividend yield on the single stock and with a volatility of σ'.

From this it follows, retracing our steps and writing the result in the original variables, that

$$V(S_1, S_2, t) = q_1 S_1 e^{-D_1(T-t)} N(d_1') - q_2 S_2 e^{-D_2(T-t)} N(d_2')$$

where

$$d_1' = \frac{\log(q_1 S_1/q_2 S_2) + (D_2 - D_1 + \frac{1}{2}\sigma'^2)(T-t)}{\sigma'\sqrt{T-t}} \quad \text{and} \quad d_2' = d_1' - \sigma'\sqrt{T-t}.$$

12.7 TWO EXAMPLES

In Figure 12.3 is shown the term sheet for 'La Tricolore' Capital-guaranteed Note. This contract pays off the *second* best performing of three currencies against the French franc, but only if the second-best performing has appreciated against the franc, otherwise

Preliminary and Indicative
For Discussion Purposes Only

'La Tricolore' Capital-guaranteed Note

Issuer	XXXX
Principal Amount	FRF 100,000,000
Issue Price	98.75%
Maturity Date	Twelve months after Issue Date
Coupon	Zero
Redemption Amount	If at least two of the following three appreciation indices, namely: $\frac{\text{USD/FRF} - 6.0750}{6.0750}$; $\frac{\text{GBP/FRF} - 10.2000}{10.2000}$; $\frac{\text{JPY/FRF} - 0.05120}{0.05120}$ are positive at Maturity, the Note will redeem in that currency whose appreciation index is the second highest of the three; in all other circumstances the Note will redeem at Par in FRF. If the Note redeems in a currency other than FRF, the amount of that currency shall be calculated by dividing the FRF Principal Amount by the spot Currency/FRF exchange rate prevailing on the Issue Date.

This indicative termsheet is neither an offer to buy or sell securities or an OTC derivative product which includes options, swaps, forwards and structured notes having similar features to OTC derivative transactions, nor a solicitation to buy or sell securities or an OTC derivative product. The proposal contained in the foregoing is not a complete description of the terms of a particular transaction and is subject to change without limitation.

Figure 12.3 Term sheet for 'La Tricolore' Capital-guaranteed Note.

it pays off at par. This contract does not have any unusual features, and has a value that can be written as a three-dimensional integral, of the form (12.2). But what would the payoff function be? You wouldn't use a partial differential equation to price this contract. Instead you would estimate the multiple integral directly by the methods of Chapter 29.

The next example, whose term sheet is shown in Figure 12.4, is of basket equity swap. This rather complex, high-dimensional contract, is for a swap of interest payment based on three-month LIBOR and the level of an index. The index is made up of the weighted average of 20 pharmaceutical stocks. To make matters even more complex, the index uses a time averaging of the stock prices.

Preliminary and Indicative
For Discussion Purposes Only

International Pharmaceutical Basket Equity Swap

Indicative terms	
Trade Date	[ ]
Initial Valuation Date	[ ]
Effective Date	[ ]
Final Valuation Date	26th September 2002
Averaging Dates	The monthly anniversaries of the Initial Valuation Date commencing 26th March 2002 and up to and including the Expiration Date
Notional Amount	US$25,000,000
Counterparty floating amounts (US$ LIBOR)	
Floating Rate Payer	[ ]
Floating Rate Index	USD-LIBOR
Designated Maturity	Three months
Spread	Minus 0.25%
Day Count Fraction	Actual/360
Floating Rate Payment Dates	Each quarterly anniversary of the Effective Date
Initial Floating Rate Index	[ ]
The Bank Fixed and Floating Amounts (Fee, Equity Option)	
Fixed Amount Payer	XXXX
Fixed Amount	1.30% of Notional Amount
Fixed Amount Payment Date	Effective Date
Basket	A basket comprising 20 stocks and constructed as described in attached Appendix
Initial Basket Level	Will be set at 100 on the Initial Valuation Date
Floating Equity Amount Payer	XXXX

Figure 12.4 Term sheet for a basket equity swap.

Floating Equity Amount	Will be calculated according to the performance of the basket of stocks in the following way: $$\text{Notional Amount} * \max\left[0, \left(\frac{\text{BASKET}_{\text{average}} - 100}{100}\right)\right]$$ where $$\text{BASKET}_{\text{average}} = 100 * \sum_{\text{20 stocks}} \left(\text{Weight} * \frac{P_{\text{average}}}{P_{\text{initial}}}\right)$$ And for each stock the Weight is given in the Appendix P_initial is the local currency price of each stock on the Initial Valuation Date P_average is the arithmetic average of the local currency price of each stock on each of the Averaging Dates
Floating Equity Amount Payment Date	Termination Date

Appendix
Each of the following stocks are equally weighted (5%):
Astra (Sweden), Glaxo Wellcome (UK), Smithkline Beecham (UK), Zeneca Group (UK), Novartis (Switzerland), Roche Holding Genus (Switzerland), Sanofi (France), Synthelabo (France), Bayer (Germany), Abbott Labs (US), Bristol Myers Squibb (US), American Home Products (US), Amgen (US), Eli Lilly (US), Medtronic (US), Merck (US), Pfizer (US), Schering-Plough (US), Sankyo (Japan), Takeda Chemical (Japan).

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Figure 12.4 (*continued*)

12.8 REALITIES OF PRICING BASKET OPTIONS

The factors that determine the ease or difficulty of pricing and hedging multi-asset options are

- existence of a closed-form solution;
- number of underlying assets, the dimensionality;
- path dependency;
- early exercise.

We have seen all of these except path dependency.

The solution technique that we use will generally be one of

- finite-difference solution of a partial differential equation;
- numerical integration;
- Monte Carlo simulation.

These methods are the subjects of later parts of the book.

12.8.1 Easy problems

If we have a closed-form solution then our work is done; we can easily find values and hedge ratios. This is provided that the solution is in terms of sufficiently simple functions for which there are spreadsheet functions or other libraries. If the contract is European with no path dependency then the solution may be of the form (12.2). If this is the case, then we often have to do the integration numerically. This is not difficult. Several methods are described in Chapter 30, including Monte Carlo integration and the use of low-discrepancy sequences.

12.8.2 Medium problems

If we have low dimensionality, lower than three or four, say, the finite-difference methods are the obvious choice. They cope well with early exercise and many path-dependent features can be incorporated, though usually at the cost of an extra dimension.

For higher dimensions, Monte Carlo simulations are good. They cope with all path-dependent features. Unfortunately, they are not very efficient for American-style early exercise.

12.8.3 Hard problems

The hardest problems to solve are those with both high dimensionality, for which we would like to use Monte Carlo simulation, and with early exercise, for which we would like to use finite-difference methods. There is currently no numerical method that copes well with such a problem.

12.9 REALITIES OF HEDGING BASKET OPTIONS

Even if we can find option values and the greeks, they are often very sensitive to the level of the correlation. But as I have said, the correlation is a very difficult quantity to measure. So the hedge ratios are very likely to be inaccurate. If we are delta hedging then we need accurate estimates of the deltas. This makes basket options very difficult to delta hedge successfully.

When we have a contract that is difficult to delta hedge we can try to reduce sensitivity to parameters, and the model, by hedging with other derivatives. This was the basis of vega hedging, mentioned in Chapter 8. We could try to use the same idea to reduce sensitivity to the correlation. Unfortunately, that is also difficult because there just aren't enough contracts traded that depend on the right correlations.

12.10 CORRELATION VERSUS COINTEGRATION

The correlations between financial quantities are notoriously unstable. One could easily argue that a theory should not be built up using parameters that are so unpredictable. I would tend to agree with this point of view. One could propose a stochastic correlation model, but that approach has its own problems.

An alternative statistical measure to correlation is **cointegration**. Very loosely speaking, two time series are cointegrated if a linear combination has constant mean and standard deviation. In other words, the two series never stray too far from one another. This is probably a more robust measure of the linkage between two financial quantities but as yet there is little derivatives theory based on the concept.

12.11 SUMMARY

The new ideas in this chapter were the multifactor, correlated random walks for assets, and Itô's lemma in higher dimensions. These are both simple concepts, and we will use them often, especially in interest rate-related topics.

FURTHER READING

- See Hamilton (1994) for further details of the measurement of correlation and cointegration.
- The first solution of the exchange option problem was by Margrabe (1978).
- For analytical results, formulæ or numerical algorithms for the pricing of some other multifactor options see Stulz (1982), Johnson (1987), Boyle *et al*. (1989), Boyle & Tse (1990), Rubinstein (1991) and Rich & Chance (1993).
- See Emanuel Derman's autobiography for discussion of quantos (Derman, 2004).
- For details of cointegration, what it means and how it works see the papers by Alexander & Johnson (1992, 1994).
- Krekel *et al*. (2004) compare different pricing methods for basket options.

EXERCISES

1. N shares follow geometric Brownian motions, i.e.

$$dS_i = \mu_i S_i \, dt + \sigma_i S_i \, dX_i,$$

for $1 \leq i \leq N$. The share price changes are correlated with correlation coefficients ρ_{ij}. Find the stochastic differential equation satisfied by a function $f(S_1, S_2, \dots, S_N)$.

2. Using tick data for at least two assets, measure the correlations between the assets using the entirety of the data. Split the data in two halves and perform the same calculations on each of the halves in turn. Are the correlation coefficients for the first half equal to those for the second? If so, do these figures match those for the whole data set?

3. Check that if we use the pricing formula for European non-path-dependent options on dividend-paying assets, but for a single asset (i.e. in one dimension), we recover the solution found in Chapter 8:

$$V(S,t) = \frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}} \int_0^\infty e^{-\left(\log(S/S')+\left(r-\frac{1}{2}\sigma^2\right)(T-t)\right)^2 \Big/ 2\sigma^2(T-t)} \mathrm{Payoff}(S') \frac{dS'}{S0'}.$$

4. Set up the following problems mathematically (i.e. what equations do they satisfy and with what boundary and final conditions?) The assets are correlated.
   (a) An option that pays the positive difference between two share prices S_1 and S_2 and which expires at time T.

(b) An option that has a call payoff with underlying S_1 and strike price E at time T only if $S_1 > S_2$ at time T.
(c) An option that has a call payoff with underlying S_1 and strike price E_1 at time T if $S_1 > S_2$ at time T and a put payoff with underlying S_2 and strike price E_2 at time T if $S_2 > S_1$ at time T.

5. What is the explicit formula for the price of a quanto which has a put payoff on the Nikkei Dow index with strike at E and which is paid in yen. $S_\$$ is the yen-dollar exchange rate and S_N is the level of the Nikkei Dow index. We assume

$$dS_\$ = \mu_\$ S_\$ \, dt + \sigma_\$ S_\$ \, dX_\$$$

and

$$dS_N = \mu_N S_N \, dt + \sigma_N S_N \, dX_N,$$

with a correlation of ρ.